



Statistical Analysis Plan

Effects of Advanced Trauma Life Support[®]
Training Compared to Standard Care on Adult
Trauma Patient Outcomes: A Stepped-Wedge
Cluster Randomised Trial

Version 0.6.0, 2026-03-04

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1 Administrative information

1.1 Study identifiers

- Protocol version 1.5.0 dated 2025-06-12
- ClinicalTrials.gov identifier: [NCT06321419](#)
- Clinical Trials Registry - India identifier: CTRI/2024/07/071336

1.2 Changelog

Once version 1.0.0 is finalised, this section will be updated with a changelog.

1.3 Contributors

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2 Trial synopsis

Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and at least 4320 patients.

Eligibility Criteria

Clusters are secondary or tertiary hospitals providing trauma care in India that admit or refer/transfer for admission at least 400 patients with trauma per year. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained.

Participants are adult trauma patients who present to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach for collection of routinely recorded data. We will obtain informed consent for collection of non-routinely recorded data, such as quality of life and disability outcomes. Patients who are unconscious or lack a legally authorized representative will be included under a waiver of informed consent. Note that consent here refers to consent to data collection.

Trial Period February 2025 to October 2029

3 Special considerations

3.1 Funding

This trial is not yet fully funded. The Trial Management Group has decided to proceed with the trial with the expectation that additional funding will be secured. The Trial Steering Committee will be informed of the funding status at each meeting. If funding is not secured, the trial will be stopped. This will likely result in an underpowered trial. The justification for this decision is that the intervention is considered standard of care in many countries and the data collection is considered minimal risk. There is therefore a very small risk of harm to patient participants, but a potential direct benefit to those patient participants who receive the intervention. The benefit-risk ratio is therefore considered to be favourable, even in the case of an underpowered trial.

4 Aim and objectives

4.1 Aim

To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

4.2 Objectives

- To estimate the effects of ATLS® training on 30-day in-hospital mortality compared to standard care.
- To assess the robustness of the main analysis results to different model specifications through sensitivity analyses.
- To estimate the effects of ATLS® training on secondary outcomes compared to standard care.
- To explore the effects of ATLS® training on subgroups of patients compared to standard care.

5 Statistical analysis

5.1 Design

This is a batched stepped-wedge cluster randomised trial, including a total of 30 clusters (see Figure 1). Each cluster is a hospital and represents the unit for randomisation, even if one or more units are trained in the same hospital. There are two intervention conditions: ATLS® training and standard care. The trial is designed and will be analysed within a superiority framework, with ATLS® training compared with standard care for the primary and secondary outcomes.

The trial will be composed of 6 batches of identical 12-period 5-sequence designs, with one cluster being assigned to each sequence of each batch¹. Each period is one month, and each cluster will be in the trial for a total of 13 months. The intervention will be implemented during a one-month transition period, which will be excluded from the analysis. There will be an anticipated overlap of 6 months between successive batches.

Clusters will be sequentially allocated to batches based on when they enter the study, and within each batch clusters will then be randomised to an intervention implementation sequence. Individual patients will be included cross-sectionally for the primary outcome, although some secondary outcomes will be assessed at multiple follow-up time points. The anticipated trial period is from February 2025 to October 2029.

Because certain secondary outcomes will be more burdensome to collect and because we anticipate larger effects on these outcomes, we will nest a staircase design within the main stepped-wedge design to measure these outcomes (see Figure 2)². The staircase design will include a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

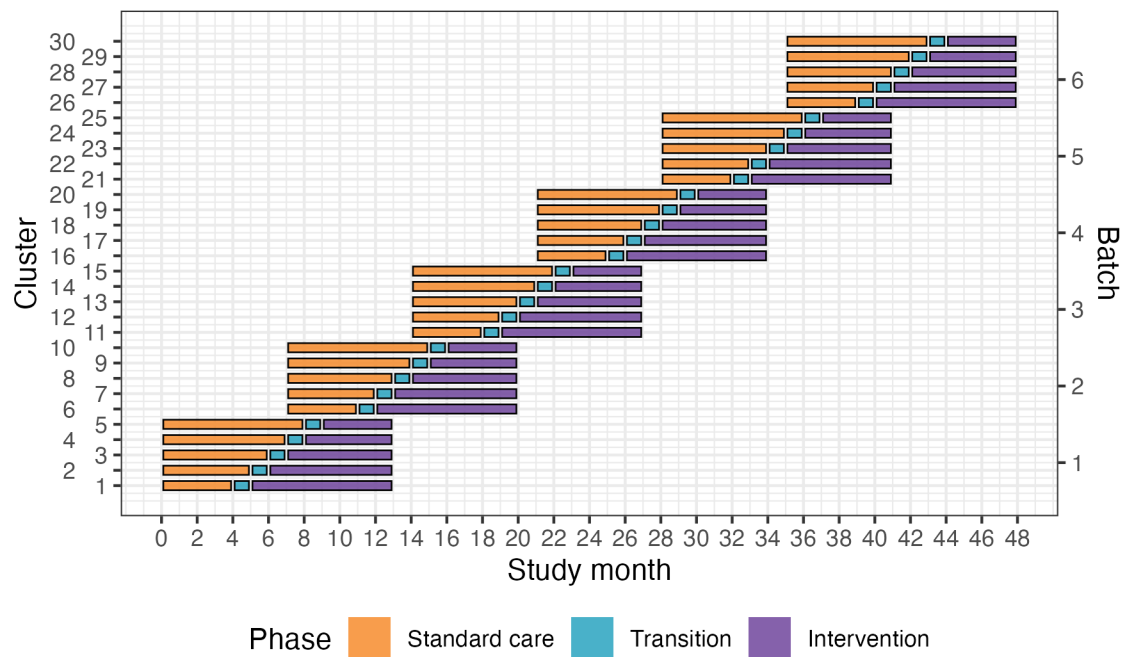


Figure 1: Trial design. Lines represent the duration of patient enrolment across clusters and phases. Clusters will be sequentially allocated to a batch based on when they enter the study. Within each batch clusters will then be randomised to an intervention implementation sequence.

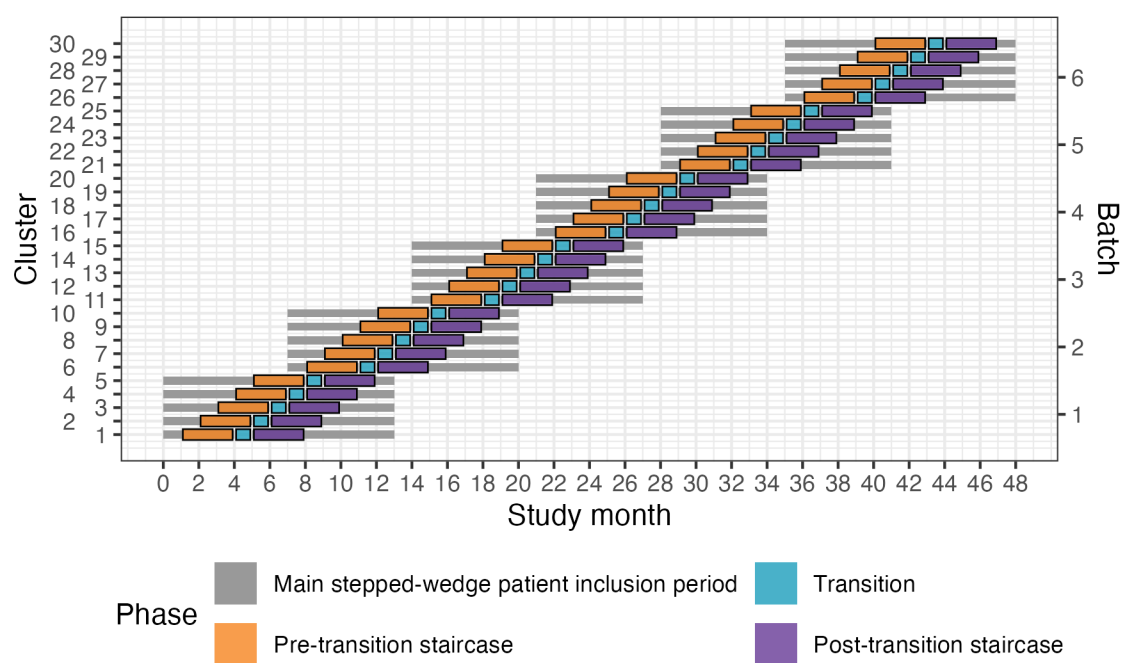


Figure 2: Nested staircase design. The design includes a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

5.2 Randomisation of clusters

We will enrol and assign clusters to batches once eligibility is confirmed and ethical approval is obtained. Each batch will include hospitals from different regions to optimise trial logistics and training delivery. Within each batch, clusters will be randomised to the intervention implementation sequences allocated to that batch.

Randomisation is conducted at the cluster level and stratified by batch. Within each batch, allocation is balanced on cluster size, defined as the monthly volume of eligible patients, using covariate-constrained randomisation (CCR). CCR was selected to improve balance between intervention conditions on this key design variable.

For example, in Batch 1, five clusters were randomised to one of five sequences using CCR. A statistician (JM) implemented a computer algorithm in R to evaluate all possible allocations ($n = 120$) and calculate a balance metric reflecting how evenly cluster size would be distributed between intervention and control conditions. For each allocation, the squared difference in anticipated cluster size between conditions was calculated; because only one covariate was used, this squared difference constituted the balance statistic.

The 10% of allocations demonstrating the greatest imbalance were excluded, and the remaining allocations formed the candidate set. The final allocation was then selected at random from this set. Owing to limitations in the algorithm, the balance score was calculated assuming a complete design, although the trial design includes transition periods. This CCR procedure is repeated independently for each batch.

The random allocation sequence is generated by JM and communicated to the trial team to allow logistical arrangements, including securing places on upcoming ATLS® training courses, to be initiated. Allocation order will be concealed from participating sites for as long as logistically feasible, recognising that advance planning is required to release physicians for training.

5.3 Outcomes

A summary of all outcomes, analysis sets, effect measures, and handling of intercurrent events is provided in Table 1.

5.3.1 Primary outcome

The primary outcome will be in-hospital mortality within 30 days of arrival at the emergency department. We define this outcome as any death occurring in the hospital during the initial admission, up to 30 days post arrival at the emergency department. Clinical research coordinators will extract information on death from patient hospital records. If the patient has been transferred to another hospital, the clinical research coordinators will collect data on this outcome by calling the patient or a patient representative, or by contacting the hospital to which the patient was transferred. Data on this outcome will be collected continuously during the trial.

5.3.2 Secondary outcomes

5.3.2.1 Collected using the main stepped-wedge design

- All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department. In contrast to the primary outcome, all-cause mortality includes all deaths regardless of setting. Data on this outcome will be collected in the same way as for the primary outcome.
- Length of emergency department stay. Data on this outcome will be collected from patient hospital records.
- Length of hospital stay. Data on this outcome will be collected from patient hospital records.
- Intensive care unit admission. Data on this outcome will be collected from patient hospital records.
- Length of intensive care unit stay. Data on this outcome will be collected from patient hospital records.
- Return to work at 30 days and three months after arrival at the emergency department. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.

5.3.2.2 Collected using the nested staircase design

- Adherence to ATLS® principles during initial patient resuscitation, up to one hour after the physician has first seen the patient. This assessment will be done using a 14 item checklist covering the key steps of the ATLS® primary survey, which was modelled based on previous work on ATLS® adherence³. We will consider completion of all 14 steps as 100% adherence. The clinical research coordinators collecting the data will be trained by the trial team to do this, prior to the start of the trial. We will collect this data by observing the care of a random sample of patients. The sampling will be designed as a nested staircase design.
- Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, measured by the official and validated translations of the EQ5D5L. This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. We will collect this data using a nested staircase design.
- Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, assessed using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. This data will also be collected using a nested staircase design.

Table 1: Summary of outcomes, analysis sets, effect measures, and handling of intercurrent events

Outcome	Design component		Data type	Analysis set	Main analysis model		Effect measure	Potential intercurrent events	Strategy for handling intercurrent events	
Primary outcome: In-hospital mortality within 30 days of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD		Loss to follow-up; withdrawal	Primary analysis set	
All-cause mortality within 24 hours of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD		Loss to follow-up; withdrawal	Primary analysis set	
All-cause mortality within 30 days of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD		Loss to follow-up; withdrawal	Primary analysis set	
All-cause mortality within three months of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD		Loss to follow-up; withdrawal	Primary analysis set	
Length of emergency department stay	Main wedge	stepped-	Time-to-event	Primary analysis set	Mixed-effects Fine–Gray model	Subdistribution HR		Death; transfer	Death event; competing primary analysis set	
Length of hospital stay	Main wedge	stepped-	Time-to-event	Primary analysis set	Mixed-effects Fine–Gray model	Subdistribution HR		Death; transfer	Death event; competing primary analysis set	
Intensive care unit (ICU) admission	Main wedge	stepped-	Dichotomous	Primary analysis set	Mixed-effects binomial (logit)	OR		Death; transfer	Primary analysis set	
Length of intensive care unit stay	Main wedge	stepped-	Time-to-event	ICU admissions, primary analysis set	Mixed-effects Fine–Gray model	Subdistribution HR		Death; transfer	Death event; competing primary analysis set	
Return to work at 30 days after arrival at the emergency department	Main wedge	stepped-	Dichotomous	Alive at 30 days, outcome relevant	Mixed-effects binomial (logit)	OR		Death; loss to follow-up; not relevant	Survivors only	
Return to work at three months after arrival at the emergency department	Main wedge	stepped-	Dichotomous	Alive at 3 months, outcome relevant	Mixed-effects binomial (logit)	OR		Death; loss to follow-up; not relevant	Survivors only	
Adherence to ATLS® principles during initial patient resuscitation	Nested staircase		Proportion	Observed nested sample	Mixed-effects beta (logit)	Logit-scale difference		Not observed; incomplete window	Observed sample	
Quality of life within seven days of discharge	Nested staircase		Ordinal EQ-5D-5L; VAS	Alive, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up	Survivors only	
Quality of life at 30 days after arrival at the emergency department	Nested staircase		Ordinal EQ-5D-5L; VAS	Alive at 30 days, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up	Survivors only	
Quality of life at three months after arrival at the emergency department	Nested staircase		Ordinal EQ-5D-5L; VAS	Alive at 3 months, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up	Survivors only	
Disability within seven days of discharge	Nested staircase		Ordinal WHO-DAS 2.0; summary score	Alive, nested sample	Ordinal mixed; linear mixed	COR; mean difference		Death; loss to follow-up	Survivors only	
Disability at 30 days after arrival at the emergency department	Nested staircase		Ordinal WHO-DAS 2.0; summary score	Alive at 30 days, nested sample	Ordinal mixed; linear mixed	COR; mean difference		Death; loss to follow-up	Survivors only	
Disability at three months after arrival at the emergency department	Nested staircase		Ordinal WHO-DAS 2.0; summary score	Alive at 3 months, nested sample	Ordinal mixed; linear mixed	COR; mean difference		Death; loss to follow-up	Survivors only	

5.4 Statistical hypotheses

The primary analysis will estimate the effect of ATLS training on 30-day in-hospital mortality compared with standard care. The treatment effect will be expressed as an odds ratio (OR) and as an absolute risk difference (ARD). An OR < 1 or ARD < 0 indicates lower mortality among patients treated in hospitals after ATLS implementation compared with standard care. Based on our pilot study and systematic review of the literature, we expect the intervention to reduce mortality.

5.5 Sample size calculations

5.5.1 Main stepped wedge design and primary outcome

With 30 clusters across 6 batches and a total sample size of 4320 our study has ~90% power across different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC) to detect a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training (see Figure 3 A)⁴. This 5% absolute reduction in mortality is a clinically meaningful and conservative estimate based on our updated systematic review and pilot study^{5,6}. We allowed for the clustered design, incorporated the one month transition period, and assumed: a discrete time decay correlation structure, an ICC of 0.02 (but considered sensitivity across the range 0.01-0.05), and a CAC of 0.9 (but considered sensitivity across the range 0.8-1.0), based on our pilot study and current guidance⁷⁻¹⁰. We included the CAC to allow for variation in clustering over time. We assume that each cluster will contribute approximately 12 observations per month to the analysis, but allowed for substantial variation in cluster sizes, based on our previous work. We also estimated the sample size required to detect a reduction in the primary outcome from 10% to 7.5%, which would require at least 30 observations per period and cluster (see Figure 3 B).

5.5.2 Nested staircase design and secondary outcomes

The secondary outcomes that will be measured using the nested staircase design are secondary outcomes, and the power calculations presented here are intended only to inform the sample size required for their data collection within the nested staircase design. These outcomes are adherence to ATLS® principles during initial patient resuscitation, quality of life, and disability. The expected effect of the intervention on each of these outcomes are an improvement in adherence from 50% during standard care to 70% after training¹¹, an increase in EQ5D5L health status from 70 during care to 75 after training¹², and finally a decrease in disability from a baseline value of 25 during standard care to 22.5 after training¹³. For quality of life and disability, these effects correspond to standardised effect sizes, expressed as Cohen's *d*, of 0.5. With 30 clusters in total, six per sequence, for a discrete time decay correlation structure, with ICCs of 0.01 to 0.15 and a CAC of 0.8, there is >80% power to detect these effects by including four patients in each cluster in each period. Accounting for loss to follow-up, we will include at least six patients per cluster per month. These patients will be a random subset of patients included in the primary outcome data collection during the staircase months. The random subset will be selected using stratified random sampling by shift, meaning that the timing of the clinical research coordinator's shift will be randomised to cover approximately eight hours during the morning, afternoon, or night shift. Adherence to ATLS® principles, quality of life, and disability will be measured in all patients included

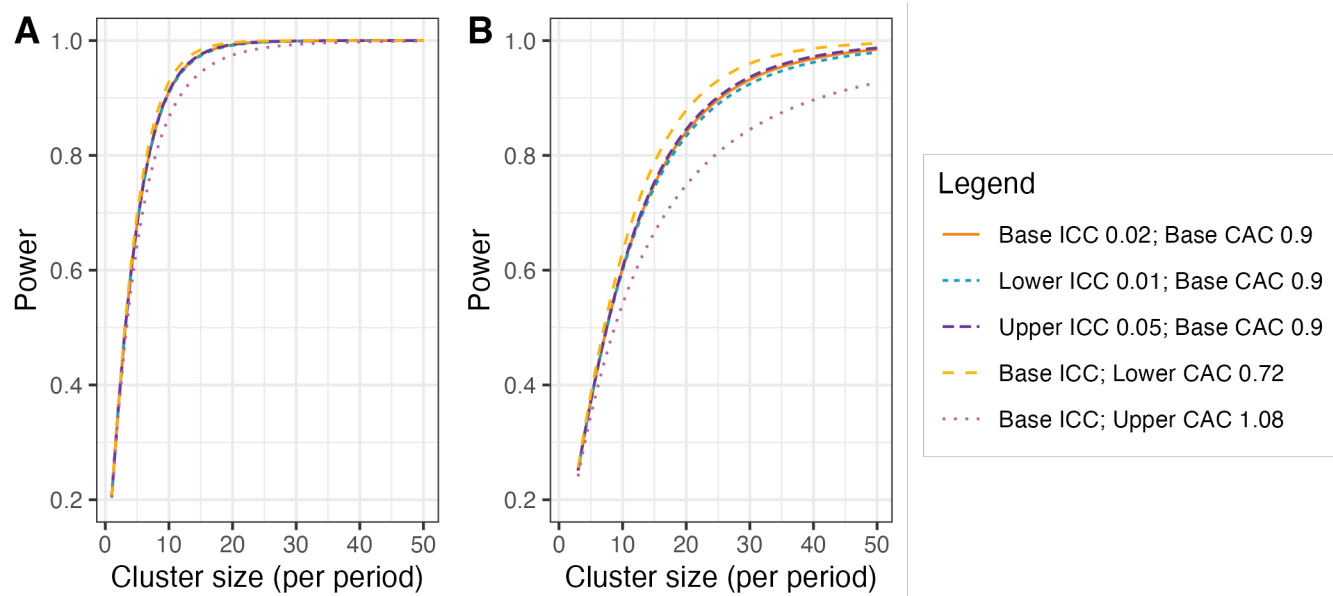


Figure 3: Power curves for different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC). **A**) Shows power curves assuming a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training. **B**) Shows power curves assuming a reduction in the primary outcome from 10% under standard care to 7.5% after ATLS® training. Under this scenario, we would need to increase the sample size per month to around 30 observations to achieve 90% power under most combinations of CAC and ICC.

during these shifts. In each hospital, the number of shifts that will be randomised will be determined by the volume of patients included during the months preceding the staircase months.

5.6 Statistical principles

5.6.1 Statistical software

We will use the R Statistical Software for all analyses¹⁴.

5.6.2 Levels of statistical significance and confidence

We will use a two-sided significance level of 0.05 for all analyses, and we will report 95% confidence intervals (CI) for all estimates. We will not adjust for multiple testing because no secondary outcome is regarded as singularly more important.

5.7 Analysis sets

The trial has a hierarchical structure in which hospitals represent the clusters, clinical units within hospitals deliver care, and individual patients constitute the unit of observation and analysis. Randomisation is performed at the hospital level, with all trained units within the same hospital treated as part of the same cluster. The primary analysis set will include all observations within clusters with available data on the primary outcome and irrespective of the receipt of the intervention.

Clusters and observations within clusters will be considered exposed to the intervention after the date at which the cluster was scheduled to transition. All data will be included with the exception of the transition phases. Group allocation for a patient depends on the period in which the patient was admitted to the hospital. If clusters deviate considerably from the scheduled date of transition, defined as more than four weeks, we will conduct a sensitivity analysis using the actual date of transition to define exposure.

The analysis set will be specified for each analysis. For the primary outcome and mortality secondary outcomes, the analysis set is all eligible patient participants included in the main stepped-wedge trial. For quality-of-life, disability, and return-to-work outcomes, the analysis set is patients alive at the relevant follow-up time point. For adherence to ATLS® principles, the analysis set is the random sample of patients observed within the nested staircase design.

The analysis set for each outcome will include all available participants from the relevant target population with observed outcome data, subject to any restrictions imposed by the outcome definition or the nested staircase sampling scheme. For adjusted analyses, the analysis set may be further restricted by the availability of covariate data, as described in Section 5.10.4.

We will use a CONSORT diagram to display the flow of hospitals, clusters and patients through the trial, see Figure 4. We will present cluster level summaries of the intervention effect. We will report the study according to the CONSORT guidelines for stepped-wedge randomised trials¹⁵.

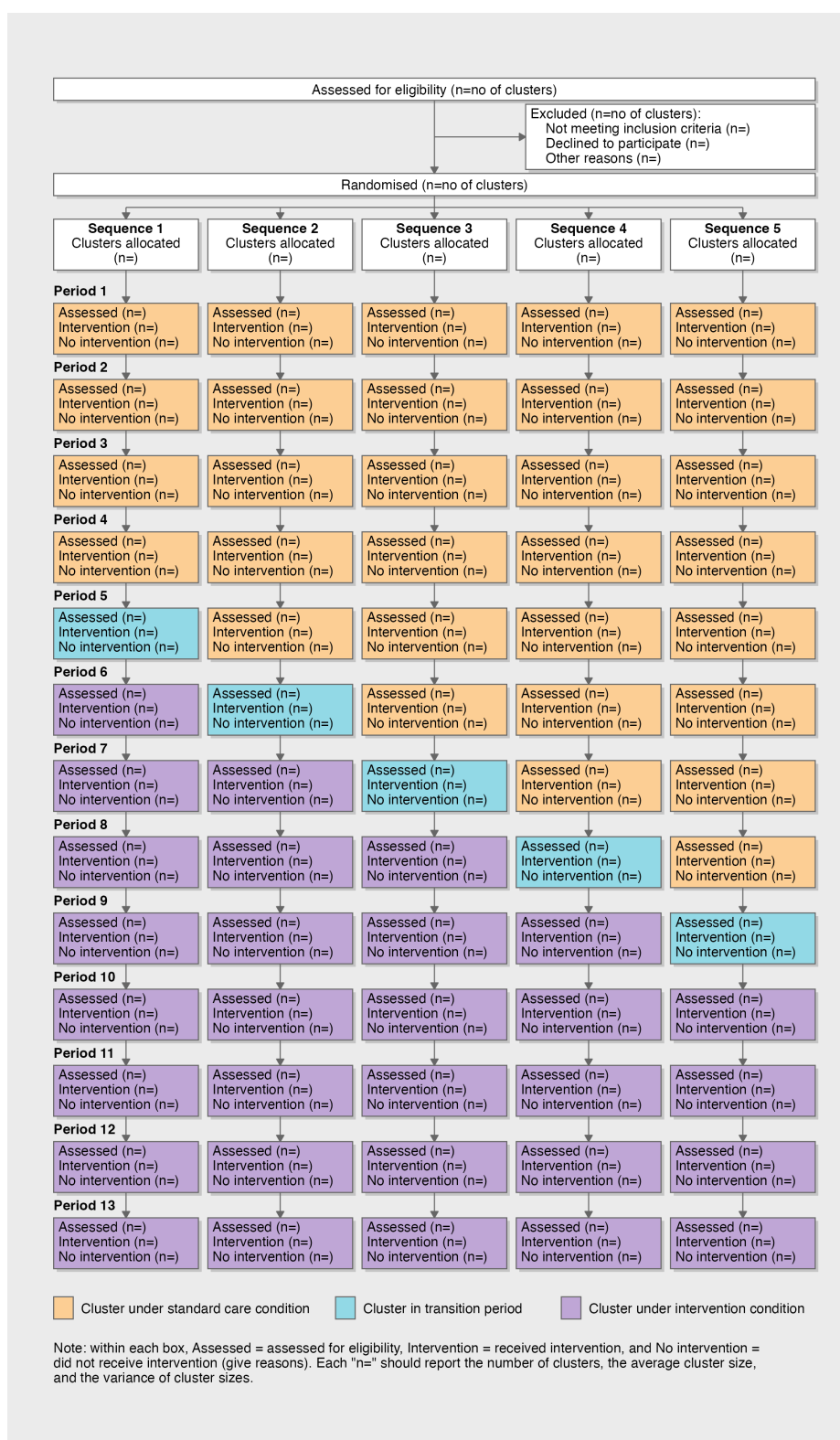


Figure 4: Consort diagram. The diagram displays the flow of clusters and participants through the trial.

Table 2: Cluster characteristics

Characteristic	Implementation sequence					Overall
	Sequence 1	Sequence 2	Sequence 3	Sequence 4	Sequence 5	
Monthly trauma patient volume, Median (Q1, Q3)						
Hospital beds, Median (Q1, Q3)						
Intensive care unit beds, Median (Q1, Q3)						
Dedicated trauma beds, Median (Q1, Q3)						
Specialities available around the clock, n (%)						
General surgery						
Orthopaedics						
Neurosurgery						
Vascular surgery						
Interventional radiology						
Emergency medicine						
Facilities available around the clock, n (%)						
X-ray						
Ultrasound/FAST						
CT						
MRI						
Blood bank						
Initial resuscitation provider, n (%)						
Casualty medical officers						
Emergency medicine residents						
Surgical residents						

5.8 Baseline analyses

5.8.1 Cluster characteristics

We will describe cluster characteristics including location and size using frequencies and percentages for discrete variables and medians and interquartile ranges (Q1-Q3) for continuous variables. See Table 2 for details.

5.8.2 Patient characteristics

We will describe patient characteristics at baseline, meaning all pre-training periods, per treatment group and overall using frequencies and percentages for discrete variables and means, standard deviations, medians and interquartile ranges (Q1-Q3) for continuous variables. We will not adjust for clustering when presenting baseline characteristics.

5.9 Interim analysis

There will be one interim analysis after half of the batches have completed the trial. The interim analysis will be assessed by the joint Trial Steering and Data Monitoring Committee. The purposes of this interim analysis will be to assess the trial's feasibility and recommend that the trial be stopped if it is not feasible (e.g. if hospitals fail to adhere to the randomisation schedule or if there are substantial missing data in outcomes), and to compare characteristics across intervention conditions to monitor for differential recruitment/ascertainment between the intervention and control groups. No efficacy stopping is planned because any estimates obtained at the time point of the interim analysis will be associated with major statistical uncertainties.

5.10 Analysis of the primary outcome

The primary outcome is in-hospital mortality within 30 days of arrival at the emergency department and will be analysed as a dichotomous variable. We will estimate the intervention effect as the odds ratio (OR) of death between the ATLS® and standard care arms, with an OR < 1 indicating lower odds of death in the ATLS® arm compared to the standard care arm and vice versa. We will also estimate the absolute risk difference (ARD) of death between the ATLS® and standard care arms, with an ARD < 0 indicating lower risk of death in the ATLS® arm compared to the standard care arm and vice versa.

We will use a mixed-effects binomial regression model to estimate the effect of ATLS® on mortality. The OR will be estimated using the logit link, and the ARD will be estimated using the identity link. The primary analysis will be based on the most complex model that converges from a prespecified sequence of models of decreasing complexity, and this model will then be subject to sensitivity analyses. The analysis will proceed from the most complex model to progressively simpler models if there are convergence or numerical stability issues. If the binomial model with the logit link converges but the corresponding model with the identity link does not, then we will report only an OR.

All models in this sequence will include a fixed effect for intervention exposure and adjust for time, although the parametrisation of time and the random effects structure may be simplified.

The model sequence is defined as follows:

1. **Primary model:** a model including fixed effects for period, modelled as a categorical variable with separate period effects for each batch, and a fixed effect for intervention exposure. Clustering will be accounted for by including a random intercept for cluster and a random cluster-by-period effect, both assumed to follow a normal distribution. The model is specified in Equation 1.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_b t + \theta X_b k t + \alpha_b k + \gamma_b k t \quad (1)$$

2. **Simplified random effects structure:** a model including fixed effects for period and intervention exposure and a random intercept for cluster only, thereby removing the random cluster-by-period effect. The model is specified in Equation 2.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_b t + \theta X_{bkt} + \alpha_b k \quad (2)$$

3. **Model with shared period effects:** a model in which period effects are shared across batches, with time adjusted for using a common period effect or calendar time modelled directly (e.g., as a continuous covariate or using splines), while retaining the random effects structure. The model is specified in Equation 3.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (3)$$

4. **Combined simplified model:** a model that includes both a simplified random effects structure (random intercept for cluster only) and shared period effects. This represents the most parsimonious mixed-effects model considered acceptable for the primary analysis. The model is specified in Equation 4.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} \quad (4)$$

Where:

- $g(\cdot)$: the link function, specified as either the logit link, $g(p) = \log\{p/(1-p)\}$, or the identity link, $g(p) = p$, depending on the estimand of interest.
- θ : **the intervention effect**, represented as a fixed effect of intervention exposure (the primary estimand); under the logit link this represents the log-odds ratio for death associated with ATLS® exposure compared with standard care, and under the identity link it represents the absolute risk difference.
- $\Pr(Y_{bkti} = 1)$: the probability of death for patient $i = 1, \dots, m$ in cluster $k = 1, \dots, 30$ during period $t = 1, \dots, 12$ in batch $b = 1, \dots, 6$.
- μ : the intercept, representing the value of the linear predictor for the reference levels of all fixed effects, with random effects equal to zero.
- β_t : the fixed effect of period t , shared across batches.
- β_{bt} : the fixed effect of period t in batch b , accounting for a separate period effect for each batch, resulting in a total of $6 \times (12)$ period effects.
- X_{bkt} : the intervention exposure indicator for cluster k during period t in batch b , with $X_{bkt} = 1$ for ATLS® and $X_{bkt} = 0$ for standard care.
- α_{bk} : a normally distributed random effect for cluster k in batch b , representing between-cluster (hospital-level) variability in the outcome.
- γ_{bkt} : a normally distributed random effect for cluster k in period t in batch b , representing within-cluster variation over time and capturing cluster-specific deviations from the average period effect.

The distributional assumptions for the random effects are specified in Equation 5.

$$\begin{aligned}
 \alpha_{bk} &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\alpha}^2), \\
 \gamma_{bkt} &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\gamma}^2), \\
 \alpha_{bk} &\perp \gamma_{bkt}.
 \end{aligned} \tag{5}$$

To correct potential inflation of the type I error rate due to the small number of clusters, a small-sample correction will be applied. Our primary method will be the Kenward–Roger correction¹⁶, but this choice may be updated based on methodological developments available closer to the end of patient inclusion. The correction applied may also differ for outcomes collected via the complete stepped-wedge design and the incomplete nested staircase designs.

All models will be fitted using residual pseudo-likelihood estimation based on linearisation with subject-specific expansion (RSPL), due to its computational stability and robust performance in small-sample settings. If the binomial model with the logit link converges but the corresponding model with the identity link does not, then we will report only an OR.

5.10.1 Sensitivity analyses

We will conduct sensitivity analyses to assess the robustness of the primary analysis results to different modelling assumptions. In all sensitivity analyses, the focus will be on the stability of the estimated intervention effect θ . Unless otherwise stated, each sensitivity analysis will be operationalised using two separate models, one with the logit link and one with the identity link.

5.10.1.1 Models with autoregressive within-cluster correlation

We will explore whether models with more complex correlation structures provide a better fit to the data. These models are not used as primary analysis models because there is limited understanding of when such models converge and how best to choose between different plausible correlation structures. Instead, we include these models as sensitivity analyses.

We will first explore whether allowing temporal correlation of outcomes within clusters affects inference on the intervention effect. Specifically, we will consider a discrete time-decay correlation structure by specifying a random cluster effect with an autoregressive structure of order one (AR(1)), described in Equation 6. This parameterisation assumes that correlations between cluster-level random effects decay geometrically with increasing temporal separation between periods.

$$\alpha_{bk,t} = \rho \alpha_{bk,t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\alpha}^2) \tag{6}$$

Where:

- ρ is the correlation parameter governing the association between adjacent periods within the same cluster.
- $\alpha_{bk,t}$ is the random effect for cluster k in period t in batch b .
- ϵ_t is a normally distributed error term.

We note that this AR(1) formulation differs slightly from alternative parameterisations used elsewhere in which correlations are specified through a joint covariance matrix. Although both approaches capture decaying temporal correlation, the parameter ρ is not directly equivalent to parameters used in other specifications and should not be interpreted interchangeably.

5.10.1.2 Models with batch-specific secular trends

To assess sensitivity to assumptions about common secular trends across batches, we will include a random batch-by-period effect allowing each batch to follow its own time trend. This specification accounts for the randomisation by batches and permits additional between-batch heterogeneity over time. The model is specified in Equation 7.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + \delta_{bt} \quad (7)$$

Where δ_{bt} represents a random effect for batch b during period t , capturing between-batch variation in secular trends over time.

5.10.1.3 Models with random cluster by intervention effects

To assess sensitivity to the assumption of a common intervention effect across clusters, we will extend the primary analysis model to allow the intervention effect to vary by cluster. Specifically, we will include a random cluster-level slope for the intervention exposure. The model is specified in Equation 8.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + u_{bk}X_{bkt} \quad (8)$$

Where:

- $u_{bk} \sim \mathcal{N}(0, \sigma_u^2)$ is a random effect representing cluster-specific deviations from the average intervention effect.
- u_{bk} is assumed to be independent of the cluster random effect α_{bk} and the cluster-by-period random effect γ_{bkt} .

5.10.1.4 Models with correlated random cluster-by-intervention effects

We will additionally explore sensitivity to the assumption that the random cluster-by-intervention effect is independent of the other random effects in the model. In this sensitivity analysis, we will fit a model that allows the random cluster-by-intervention effect u_{bk} to be correlated with both the cluster random intercept α_{bk} and the cluster-by-period random effect γ_{bkt} . This relaxes the independence assumptions of the random-effects structure and allows for potential associations between baseline cluster risk, cluster-specific time deviations, and cluster-specific intervention effects. The model is specified in Equation 9.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + u_{bk}X_{bkt} \quad (9)$$

We assume a joint multivariate normal distribution for the random effects:

$$\begin{pmatrix} \alpha_{bk} \\ \gamma_{bkt} \\ u_{bk} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{\alpha}^2 & \sigma_{\alpha\gamma} & \sigma_{\alpha u} \\ \sigma_{\alpha\gamma} & \sigma_{\gamma}^2 & \sigma_{\gamma u} \\ \sigma_{\alpha u} & \sigma_{\gamma u} & \sigma_u^2 \end{pmatrix} \right).$$

5.10.1.5 Models with time modelled with a spline function

We will further explore the potential for a time-varying treatment effect¹⁷. To explore if the fixed period effect is both parsimonious and adequate to represent the extent of any underlying secular trend, we will model calendar time using a shared natural cubic spline basis across all batches with knots at the equally spaced time points 3, 6 and 9. This will result in five spline basis functions, because the natural cubic splines are modelled with three degrees of freedom but are constrained to be linear before the first and after the last knot. The model is specified in Equation 10.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \sum_{j=1}^5 \beta_j S_j(t, \{3, 6, 9\}) + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (10)$$

Where:

- t denotes calendar time (period index) common to all batches.
- $S_j(t, \{3, 6, 9\})$ denotes the j th natural cubic spline basis function, shared across batches.
- β_j is the coefficient for the j th spline basis function.

5.10.1.6 Models exploring lag and weaning effects

We will also extend the primary analysis model to include an interaction between treatment and number of periods since first treated, to examine if there is any indication of a relationship between duration of exposure to the intervention and outcomes. This will allow us to model different lag effects (whereby it takes time for the intervention to become embedded within the culture before its impact can properly start to be realised); as well as weaning effects (whereby the effect of the intervention starts to decrease – or fade). This type of analysis attempts to disentangle how some clusters end up having a long exposure to the intervention and others have a much shorter exposure time. The model is specified in Equation 11.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \theta_{\text{int}} X_{bkt} \times T_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (11)$$

Where θ_{int} represents the interaction between intervention exposure and time since first treated, and T_{bkt} is the number of periods since first exposure.

5.10.2 Adjusted analyses

Fully adjusted covariate analysis will additionally adjust for (expected % missing data in parentheses):

- Age (<5%)
- Sex (<5%)
- Systolic blood pressure (25%)
- Glasgow Coma Scale (25%)
- Injury Severity Score (10%)
- Mechanism of injury (10%)

These are known individual-level prognostic factors for the primary outcome. These covariates will be included in the models specified in Equation 1 as fixed effects. We will model the continuous covariates assuming linear effects. We will assume no interaction between subgroups and secular time trends, which may be evaluated in future analyses if indicated.

5.10.3 Subgroup analyses

We will perform the following subgroup analyses:

- geographical region, defined using the state in which the participating hospital is located;
- age groups, defined as older adolescents (15-19 years), young adults (20-24 years), adults (25-59 years), and older adults (60 years and older)¹⁸
- sex, using the levels male and female;
- clinical cohorts, defined as blunt multisystem trauma, penetrating trauma, and severe isolated traumatic brain injury, with modification to avoid overlap between the cohorts¹⁹;
- patients with major trauma, defined as patients with an injury severity score (ISS) of 16 or higher;
- cluster size (number of patients per hospital)

These subgroup analyses will be conducted by adding the subgroup variable and the interaction between the subgroup variable and the intervention exposure variable as fixed effects to the models specified in Equation 1.

5.10.4 Treatment of missing data

We will present the frequency and percentage of missing data for all variables. If the percentage of missing data for the primary outcome is less than 10%, we will perform a complete case analysis. If the percentage of missing data for the primary outcome is 10% or more, we will handle missing data depending on the missing data mechanism. If the data are missing at random (MAR), we will perform multiple imputation using multiple imputation by chained equations (MICE), imputing data for the primary outcome as well as all covariates included in the fully adjusted model. The number of imputations will be determined by the percentage of missing data, with a minimum of 20 imputations. If there is evidence that the data are missing not at random (MNAR), we will explore the impact of this assumption using a sensitivity analysis (e.g., pattern mixture models or selection models) to assess how robust our findings are to different assumptions about the missing data mechanism. Evidence that data are MNAR may include systematic differences in missingness across observed covariates, patterns

suggestive of non-random dropout, or clinical expectations that missingness is related to unmeasured outcomes. Additionally, we will perform diagnostic checks after multiple imputation to ensure the quality of the imputation process, including comparing distributions of observed and imputed data and checking convergence.

5.11 Analysis of secondary outcomes

Unless otherwise stated, secondary outcomes will be analysed using the same mixed-effects modelling framework as specified for the primary outcome, including fixed effects for time and intervention exposure and random effects for cluster and cluster-by-period, with the link function and outcome distribution adapted to the scale of the outcome.

5.11.1 All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department

These outcomes will be analysed as dichotomous variables using the mixed-effects binomial model with the logit and identity links specified in Equation 1, to estimate odds ratios and absolute risk differences.

5.11.2 Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, among survivors

Quality of life will be measured using the official and validated translations of the EQ5D5L. This tool assesses five dimensions of health-related quality of life: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension is rated on a Likert scale from 1 to 5. There is also a visual analogue scale (VAS) ranging from 0 to 100.

For each of the five ordinal dimensions, we will use a mixed-effects cumulative logit (proportional odds) model:

$$\text{logit}\{\Pr(Y_{bkti} \leq j)\} = \mu_j + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (12)$$

Where μ_j is the intercept for the j th category.

The VAS will be analysed as a continuous outcome using a linear mixed-effects model:

$$\text{VAS}_{bkti} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + \epsilon_{bkti}, \quad \epsilon_{bkti} \sim \mathcal{N}(0, \sigma^2) \quad (13)$$

5.11.3 Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, among survivors

We will measure disability using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0)²⁰. This tool assesses six domains of functioning: cognition, mobility, self-care, getting along, life activities, and participation. Each domain is rated on a likert scale from 1 to 5, with 1 indicating no difficulties and 5 indicating extreme difficulties. We will analyse each domain separately using a mixed effects ordinal model as specified in Equation 12. We will also calculate a WHODAS 2.0 summary score using the method referred to as the “complex scoring” method. This method involves summing the item scores within each of the six domains, then summing the scores of all domains, and finally transforming the total score to a 0-100 scale. We will analyse the summary score using a linear mixed effects model as specified in Equation 13.

5.11.4 Return to work at 30 days and three months after arrival at the emergency department, among survivors

Return to work will be analysed as a dichotomous outcome using the mixed-effects binomial model specified in Equation 1 with the logit link.

5.11.5 Length of emergency department stay

Length of emergency department stay will be analysed as a time-to-event outcome, defined as time from arrival at the emergency department to discharge. We will treat death as a competing event and therefore use a mixed-effects competing risks model, specifically a Fine and Gray subdistribution hazards model. The model is specified in Equation 14.

$$h_{bk}^{(s)}(t) = h_0^{(s)}(t) \exp\{\theta X_{bkt} + \beta_{bt} + \alpha_{bk}\} \quad (14)$$

Where:

- $h_{bk}^{(s)}(t)$: the subdistribution hazard for discharge at time t for patients in cluster k in batch b .
- $h_0^{(s)}(t)$: the baseline subdistribution hazard function.
- θ : under the Fine–Gray model, represents the log subdistribution hazard ratio for discharge associated with ATLS® exposure.
- All other terms are defined as in the primary analysis model.

5.11.6 Length of hospital stay

Length of hospital stay will also be analysed as a time-to-event outcome, defined as time from arrival at the emergency department to hospital discharge, using the model specified in Equation 14.

5.11.7 Intensive care unit admission

Intensive care unit admission will be analysed as a dichotomous outcome using the mixed-effects binomial model specified in Equation 1 with the logit link.

5.11.8 Length of intensive care unit stay

Length of intensive care unit stay will also be analysed as a time-to-event outcome, defined as time from admission to the intensive care unit to discharge from the intensive care unit, using the model specified in Equation 14.

5.11.9 Adherence to ATLS® principles during initial patient resuscitation

We will assess adherence to ATLS® principles using a 14-item checklist that covers the key steps of the ATLS® primary survey. Adherence will be measured as the proportion of completed items on this checklist. This outcome will be analysed using a mixed-effects beta regression model with a logit link:

We will assess adherence to ATLS® principles using a 14-item checklist that covers the key steps of the ATLS® primary survey. Adherence will be measured as the proportion of completed items on this checklist. This outcome will be analysed using a mixed-effects beta regression model with a logit link:

$$g\{\mathbb{E}(Y_{bkti})\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (15)$$

Where $\mathbb{E}(Y_{bkti})$ denotes the expected adherence proportion. The intervention effect θ represents the difference in expected adherence proportion on the logit (link) scale. Because the observed adherence proportion may take values of 0 or 1, values will be rescaled to lie in the open interval (0, 1) prior to fitting the beta regression model. A beta regression precision (dispersion) parameter will be estimated as part of the model.

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